

RQ1/RQ2 Profile Construction

Complete 108-run analysis

3 sites × 4 models × 3 retrieval strategies × 3 repetitions

Frozen login measurements, pilot results, and documentation corpora.
Tokens exclude the one-time corpus-discovery cost. Extraction latency covers context selection, model calls, and evidence validation.
Run-to-run SD is pooled within each three-repetition configuration.

Key observations

- Precision is 0.957–0.992; field accuracy is 0.783–0.910 by site.
- All 5,592 accepted citations pass the mechanical integrity checks.
- BM25 has the highest retrieval-level accuracy and lowest token cost.
- Full corpus uses 2.18× BM25 tokens without improving field accuracy.
- Gemini 3.6 Flash is most accurate; GPT-5.6 Terra is cheaper and faster.

Table 1: Results by site (RQ1)

Site	Precision	Recall	Abstention Accuracy	Field Accuracy	Citation Integrity
Anvil	0.979	0.874	0.969	0.895	1.000
Stampede3	0.957	0.904	0.934	0.910	1.000
Notre Dame CRC	0.992	0.771	0.986	0.783	1.000

Table 3: Results by model (RQ2)

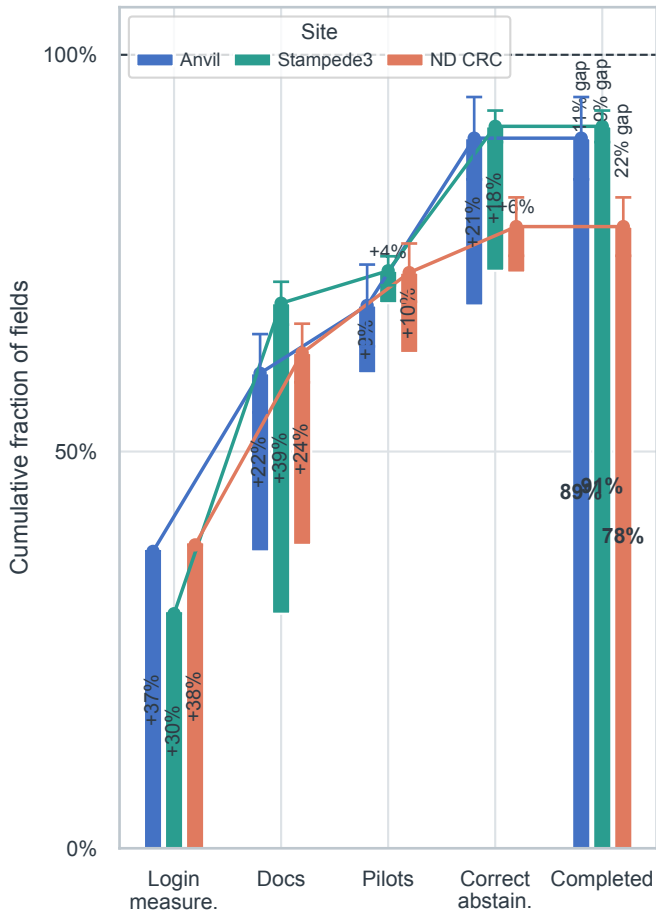
Model	Field Accuracy	Precision	Recall	Tokens	Latency	Run-to-Run SD
GPT-5 mini	0.841	0.959	0.826	89,513	362.0	0.040
GPT-5.6 Terra	0.866	0.982	0.851	58,368	51.9	0.012
Gemini 3.6 Flash	0.875	0.980	0.864	87,109	150.2	0.013
Gemini 3.1 Pro	0.869	0.980	0.857	86,481	221.3	0.014

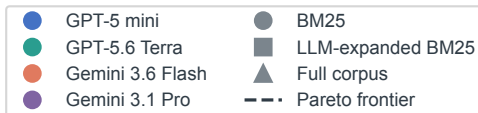
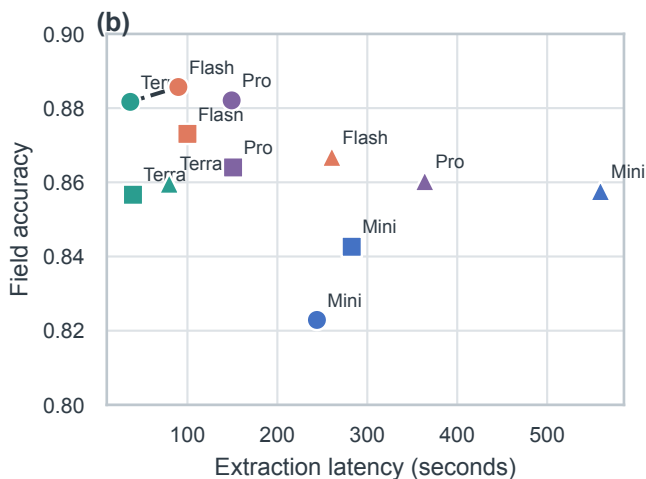
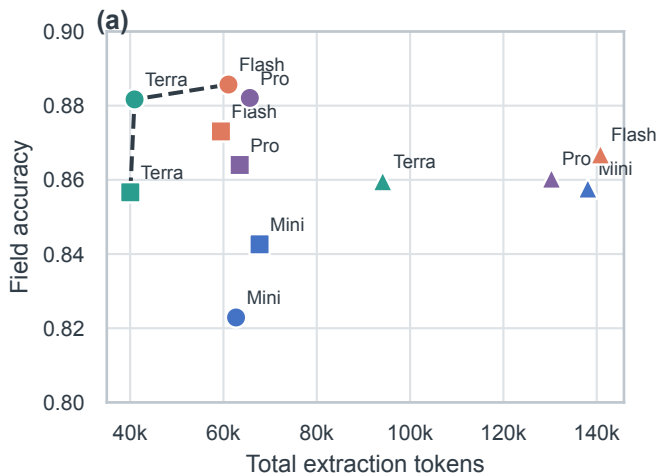
Table 4: Results by retrieval strategy (RQ2)

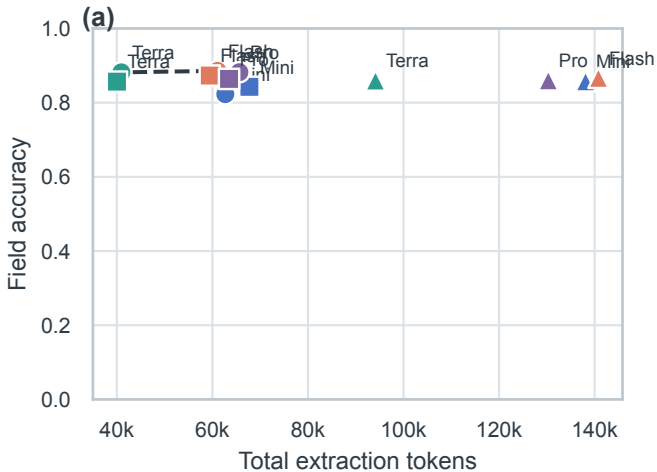
Retrieval	Field Accuracy	Precision	Recall	Tokens	Latency	Run-to-Run SD
BM25	0.868	0.976	0.855	57,590	129.9	0.021
LLM-expanded BM25	0.859	0.975	0.847	57,678	143.2	0.028
Full corpus	0.861	0.976	0.847	125,835	315.9	0.017

Table 5: Model-retrieval combinations

Model	Retrieval	Field Accuracy	Tokens	Latency	Run-to-Run SD
GPT-5 mini	BM25	0.823	62,696	244.0	0.041
GPT-5 mini	LLM-expanded BM25	0.843	67,742	282.7	0.044
GPT-5 mini	Full corpus	0.858	138,100	559.2	0.033
GPT-5.6 Terra	BM25	0.882	40,943	36.5	0.002
GPT-5.6 Terra	LLM-expanded BM25	0.857	40,027	39.4	0.021
GPT-5.6 Terra	Full corpus	0.860	94,133	79.7	0.000
Gemini 3.6 Flash	BM25	0.886	61,069	89.9	0.003
Gemini 3.6 Flash	LLM-expanded BM25	0.873	59,458	100.0	0.020
Gemini 3.6 Flash	Full corpus	0.867	140,801	260.6	0.011
Gemini 3.1 Pro	BM25	0.882	65,653	149.2	0.013
Gemini 3.1 Pro	LLM-expanded BM25	0.864	63,484	150.8	0.020
Gemini 3.1 Pro	Full corpus	0.860	130,306	363.9	0.002





(a)**(b)**